

CSA 1220 COMPUTER ARCHITECTURE

CO 1 AT-1 SET-4

Question no1:

Title:

Study of Register Operations in 8085 Microprocessor Using GNUSim8085

Aim:

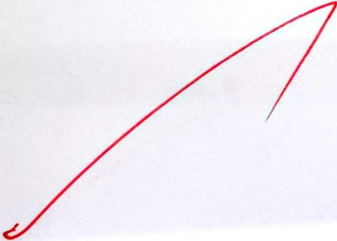
To write and execute an 8085 assembly language program that performs data transfer and arithmetic operations using multiple registers and to observe the contents of the Accumulator (A) and general-purpose registers (B, C, D, E, H and L) after each instruction.

Software Required:

- GNUSim8085 Simulator
- Windows Computer

Program:

MVI B,25H
MVI C,15H
MOV A,B
ADD C
MOV D,A
MVI E,05H
ADD E
MOV H,A
MOV L,C
HLT

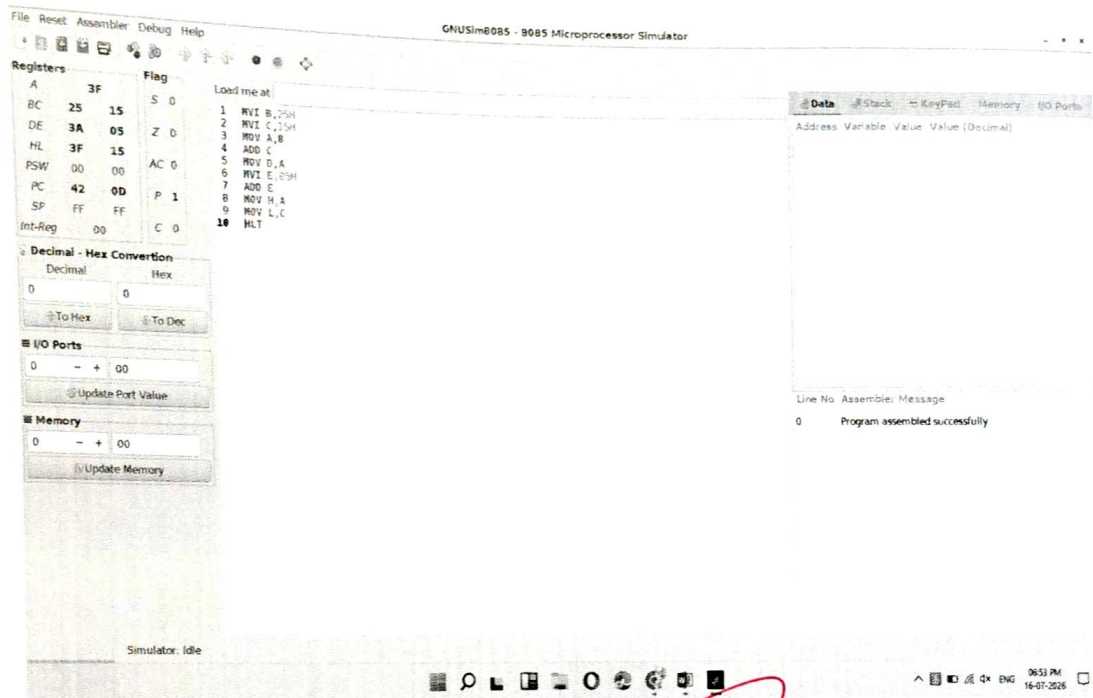


Procedure:

1. Open GNUSim8085.
2. Create a new program.
3. Enter the above assembly program.
4. Assemble the program.

5. Execute the program using Step Execution.
6. Observe the changes in registers A, B, C, D, E, H and L after every instruction.
Record the final register values.

Program output:



Result:

The program was executed successfully using **GNUSim8085**. Data transfer and arithmetic operations were performed using the general-purpose registers and the Accumulator. The register values changed after each instruction as expected. The Accumulator stored the arithmetic results, while registers B, C, D, E, H, and L were used for data storage and transfer, demonstrating the role of each register during program execution.

Question no:2

Title:

Analysis of CPU, Memory, Disk and Network Activity During Multitasking

Aim:

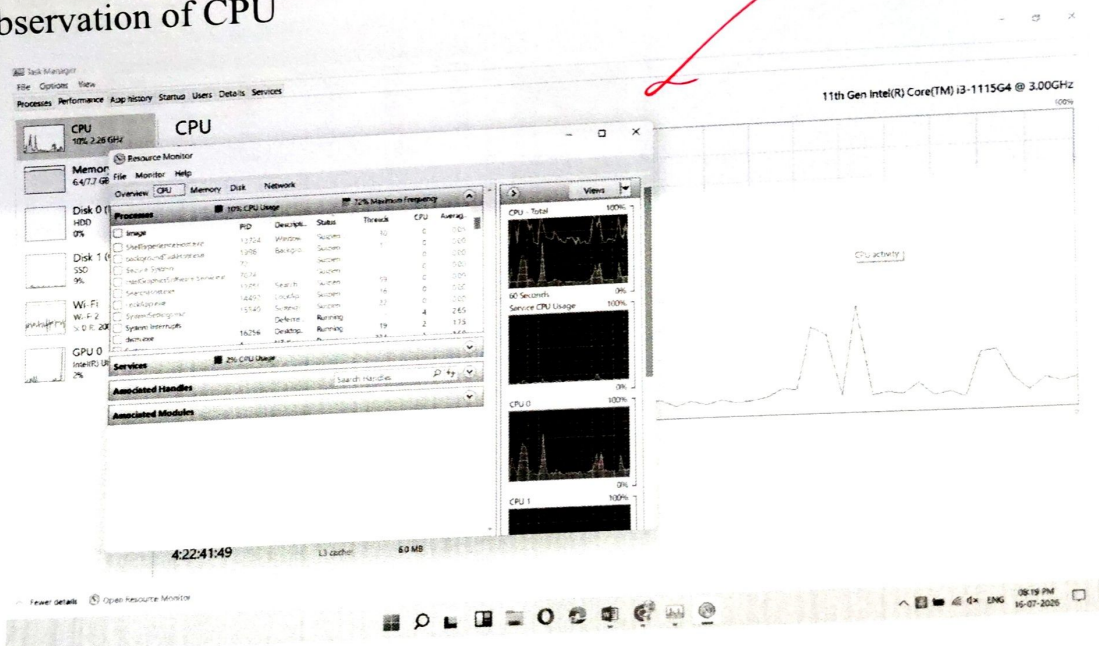
To observe the CPU, Memory, Disk, and Network usage while attending an online discussion, editing a document, and downloading files simultaneously, and to identify the hardware resource that becomes the performance bottleneck during multitasking.

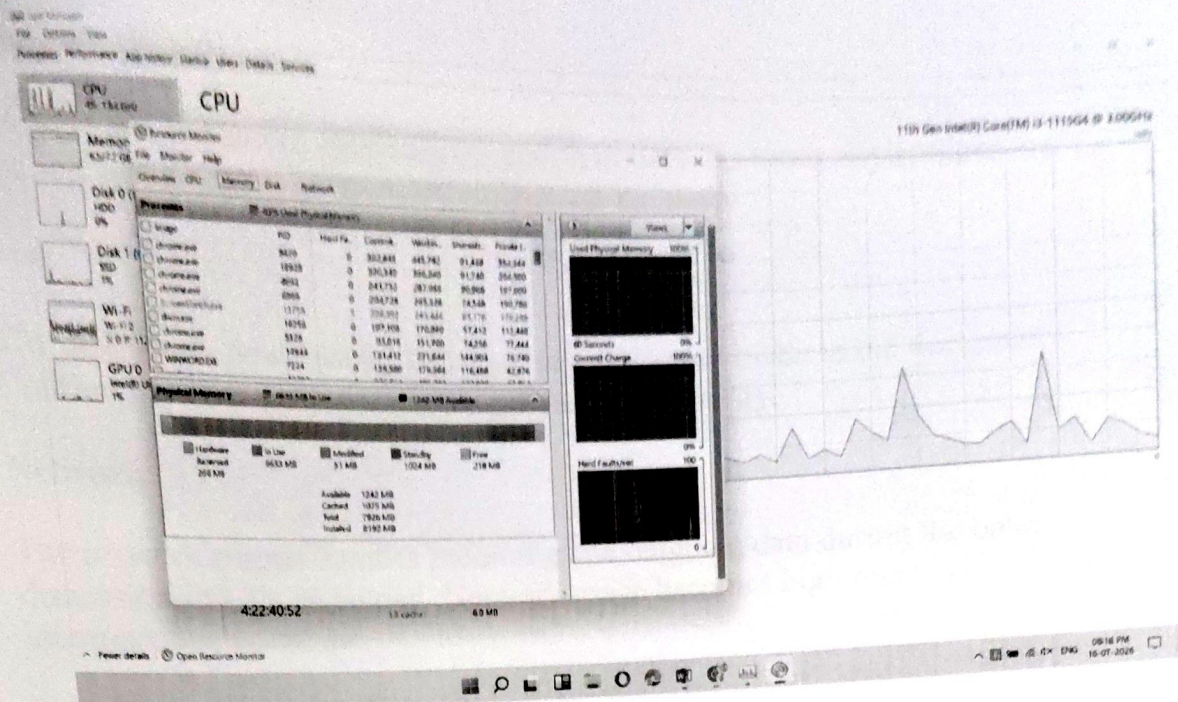
Software Required:

1. Windows Task Manager
2. Resource Monitor
3. Performance Monitor
4. Google Chrome / Microsoft Edge
5. Microsoft Word

Program:

Observation of CPU





Observation of network



Analysis

CPU

The CPU executes instructions from all running applications by rapidly switching between processes. During multitasking, the CPU schedules browser, document editing, and download tasks efficiently.

Memory (RAM)

RAM stores the active applications and their data. As more applications are opened simultaneously, memory utilization increases.

Disk

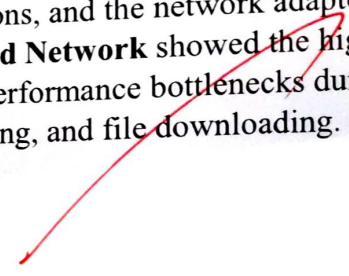
The disk stores downloaded files and saves changes made to the document. Continuous read/write operations increase disk activity.

Network

The network adapter handles incoming and outgoing data during the online discussion and file download. Network usage becomes high due to simultaneous internet communication.

Result:

The experiment was performed successfully. During multitasking, the CPU coordinated the execution of multiple applications, RAM stored active programs, the disk handled file storage operations, and the network adapter managed internet communication. The **Disk and Network** showed the highest utilization and were identified as the primary performance bottlenecks during simultaneous online discussion, document editing, and file downloading.



Question no:3

Title:

Comparison of Hardware Specifications of Two Computer Systems

Aim:

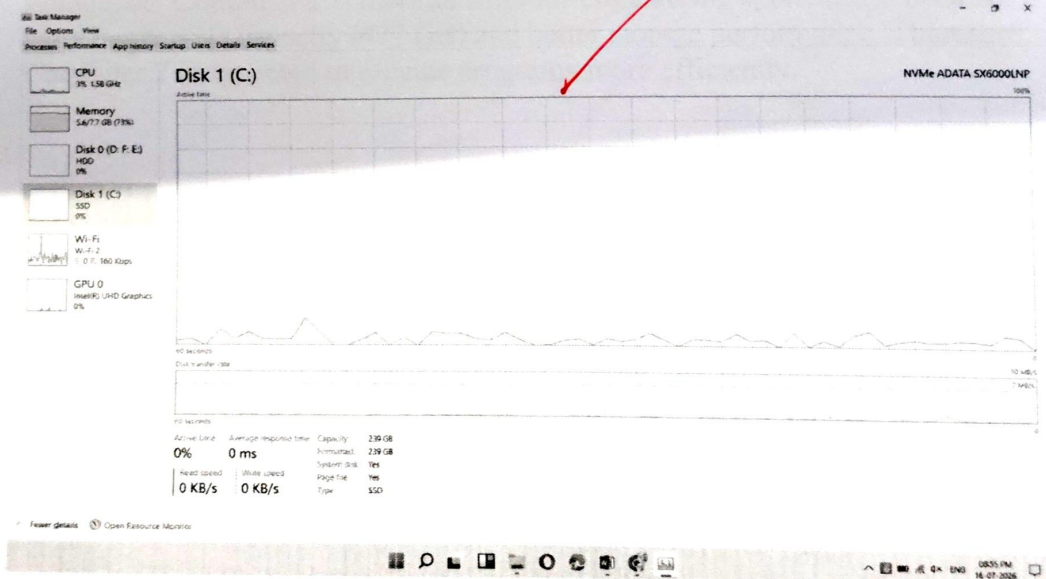
To compare the hardware specifications of two computer systems by observing the processor generation, base speed, current speed, virtualization status, installed RAM, cache memory, and storage type, and to determine which system is more suitable for running engineering applications.

Software Required

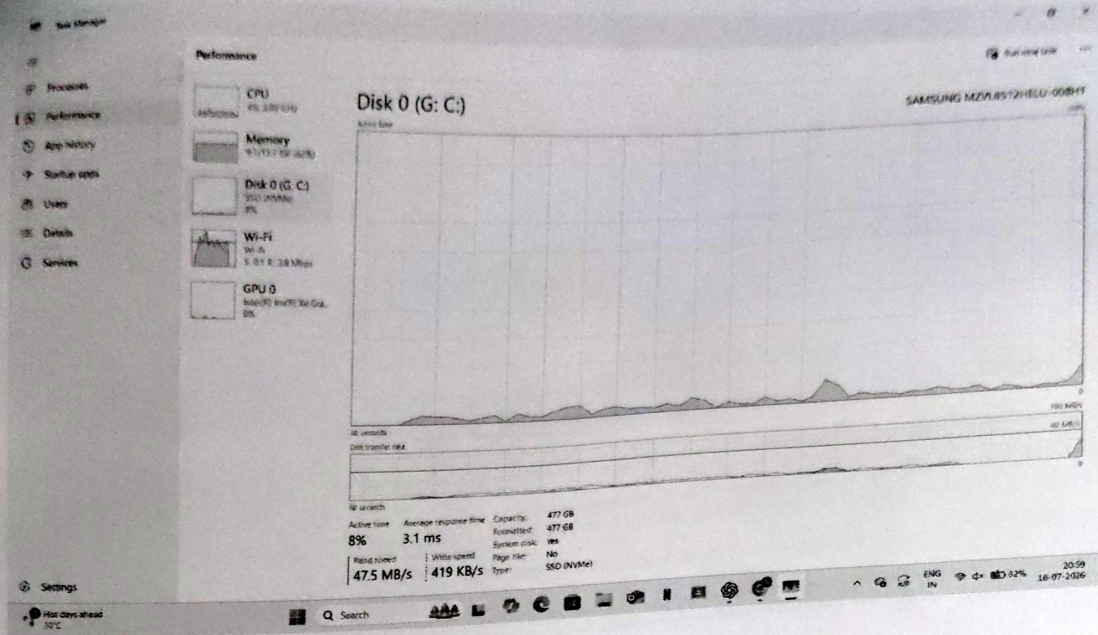
- Windows Settings
- Task Manager

Output:

System 1



System 2



Result:

The comparison was completed successfully. Both computers use NVMe SSD storage, but Computer 2 is more suitable for engineering applications because it has a larger SSD capacity (477 GB) and better storage performance. Therefore, Computer 2 is expected to execute programs more efficiently.